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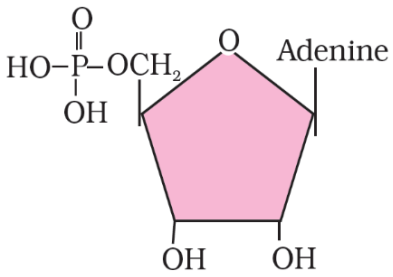
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BIOLOGY

1) Polypeptide synthesis require energy during :-
P – An amino acid activation
Q – Binding of large ribosomal subunit
R – Binding of tRNA to P site
S – Binding of tRNA to A site

(1) P, Q, R, S
(2) P, Q, R
(3) P, R, S
(4) Only P, S

2) Given below structure represent :-



(1) Nucleotide of DNA
(2) Nucleotide of RNA
(3) Nucleoside of DNA
(4) Nucleoside of RNA

3) Which of the following is not a nucleoside of RNA?

(1) Cytidine
(2) Uridine
(3) Guanosine
(4) Deoxyguanosine

4)

	Organism		**No. of Nucleotide and Base Pair**
(i)	$\phi \times 174$	A	4.6×10^6 bp
(ii)	Lambda bacteriophage	B	3.3×10^9 bp
(iii)	E.Coli	C	5386 Nucleotides
(iv)	Human	D	48502 bp

Following match is correct

(1) i = D, ii = C, iii = B, iv = A

(2) i = B, ii = D, iii = A, iv = B

(3) i = A, ii = D, iii = C, iv = B

(4) i = C, ii = D, iii = A, iv = B

5) What is the major hallmark of double helix modes of DNA given by Watson and Crick :-

(1) Base pairing between two stands.

(2) Presense of phosphodiester bond.

(3) Acidic nature of DNA.

(4) Presence of pentose sugar.

6) The successive nucleotides of DNA in a strand are linked together by :-

(1) Peptide Bond

(2) Disulphide Bond

(3) Glycosidic Bond

(4) Phosphodiester bond

7) Lac operon is turned on when allolactose binds to :-

(1) Operator gene

(2) mRNA

(3) Repressor protein

(4) Promoter site

8) On an average human gene consists of :-

(1) 3000 bases

(2) 1.4 million bases

(3) 2.4 million bases

(4) 29 bases

9) Which of the following is not the feature of human genome?

(1) Less than 2 percent of the genome code for protein

(2) Chromosome 1 has fewest gene (231)

(3) Repetitive sequences make up very large portion of human genome

(4) The functions are unknown for over 50% of the discovered genes

10) A m-RNA also has some additional sequences that are not translated called UTR. The function of UTR is:-

(1) Charging of t-RNA

(2) Formation of peptide bond

(3) Helps in efficient translation

(4) Helps in translocation

11) Polyploidy differs from aneuploidy in which of the following ways?

(1) In polyploidy, entire sets of chromosomes are increased, while in aneuploidy only one or few chromosomes increase or decrease

(2) Polyploidy is more common in plants, while aneuploidy is more common in humans

(3) Triploidy (3n) and Tetraploidy (4n) are examples of polyploidy

(4) All of the above

12) Match the following with respect to Transcription in Prokaryotes:

	Column I		**Column II**
1.	σ (Sigma) factor	(a)	Helps in promoter recognition and initiation
2.	Template strand	(b)	Serves as actual template for RNA synthesis
3.	Coding strand	(c)	Same sequence as RNA (except T→U)
4.	RNA Polymerase	(d)	Catalyzes formation of RNA

(1) 1-a, 2-b, 3-c, 4-d,

(2) 1-d, 2-b, 3-c, 4-a,

(3) 1-c, 2-d, 3-a, 4-b,

(4) 1-c, 2-d, 3-a, 4-b,

13) In prokaryotic translation, when a stop codon (UAA/UAG / UGA) enters the A-site of the ribosome, what happens?

(1) A special tRNA carrying no amino acid binds to stop codon and terminates translation

(2) Release factors recognize the stop codon and hydrolyze the bond between polypeptide and tRNA

(3) Ribosome directly dissociates without any protein involvement

(4) EF-G protein removes the last tRNA without GTP hydrolysis

14) Find out the correct statements from the followings -
(a) In lac-operon structural genes is regulated by a common promoter and regulatory gene
(b) In lac-operon one regulatory gene (the i gene) is present, here the term i refers to the inducer
(c) Lactose is the substrate for the enzyme betagalactosidase and it regulates switching on and off the operon
(d) The y-gene of lac-operon codes for permease
(e) The z-gene codes for beta-galactosidase

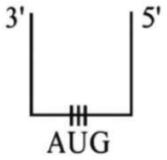
(1) a and c only

(2) a, b and e only

(3) a, c and e only

(4) a, c, d and e

15) Which of the following tRNA doesn't exist:

(1) 

(2) 

(3) 

(4) 

16) Identify the incorrect match:

(1) Ribozyme - Nucleic acid

(2) Spliceosome - Bacteria

(3) Genetic code - Triplet

(4) Replication - DNA Polymerase

17) Select the nitrogenous base(s) which is/are not common between DNA and RNA
(a) Thymine (b) Guanine (c) Adenine
(d) Uracil (e) Cytosine

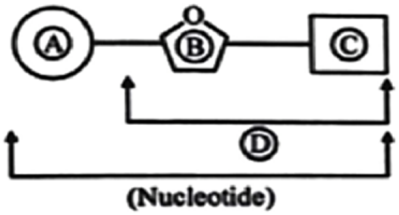
(1) only d

(2) b, c and e

(3) a and d

(4) a only

18) Identify the missing words A, B, C and D in the following figure related to DNA structure :



	A	**B**	**C**	**D**
(1)	Phosphate	Deoxyribose	Purine	Nucleoside

(2)	Nucleoside	Phosphate	Deoxyribose	Purine
(3)	Phosphate	Purine	Deoxyribose	Nucleoside
(4)	Deoxyribose	Nucleoside	Purine	Phosphate

(1) 1

(2) 2

(3) 3

(4) 4

19) Given below are some steps of transcription -
A. RNA Polymerase uses nucleoside triphosphates as substrate during elongation.
B. RNA Polymerase associates transiently with termination-factor (ρ) to terminate the transcription.
C. RNA Polymerase facilitates opening of the DNA helix.
D. Only a short stretch of RNA remains bound to the enzyme.
E. RNA polymerase binds to promoter by the help of sigma factor.
Choose the **correct sequence** of steps from the options given below :-

(1) C-E-D-A-B

(2) E-B-D-A-C

(3) B-D-E-A-C

(4) E-C-A-D-B

20) Read the following statements.
I. A purine is heterocyclic, 9-membered double ring structure with nitrogen at 1st, 3rd, 7th and 9th positions.
II. A pyrimidine is heterocyclic, 6-membered single ring structure with nitrogen at 1st and 3rd positions.
III. Purine nucleosides have 1′-9 glycosidic linkage whereas pyrimidine nucleosides have 1′-1-glycosidic linkage.
IV. Two nucleosides are linked by 3′-5′ phosphodiester linkage to form a dinucleotide.
V. Ribose sugar can be represented as $C_5H_{10}O_4$ whereas deoxyribose sugar can be represented as $C_5H_{10}O_5$.
Which of the above statement(s) is/are **correct**?

(1) Only I

(2) I, II and III

(3) III, IV and V

(4) IV and V

21) Which of the following statements about Griffith’s experiment are correct?
I. S-strain have mucus (polysaccharide) coat.
II. S-strain are virulent and cause pneumonia infection, while R-strain do not.
III. Transforming principle is associated with genetic material of R-strain.
IV. Transformation of R-strain into S-strain can take place in a test tube.

(1) I and III

(2) III and IV

(3) I, II and IV

(4) II, III and IV

22) Stability of DNA is due to.

(1) Deoxyribose sugar

(2) Presence of thymine in place of uracil

(3) Both (1) and (2)

(4) None of the above

23) If the sequence of bases in the coding strand of a doube stranded DNA is 5' GTTCGAGTC-3' the sequence bases in its mRNA will be.

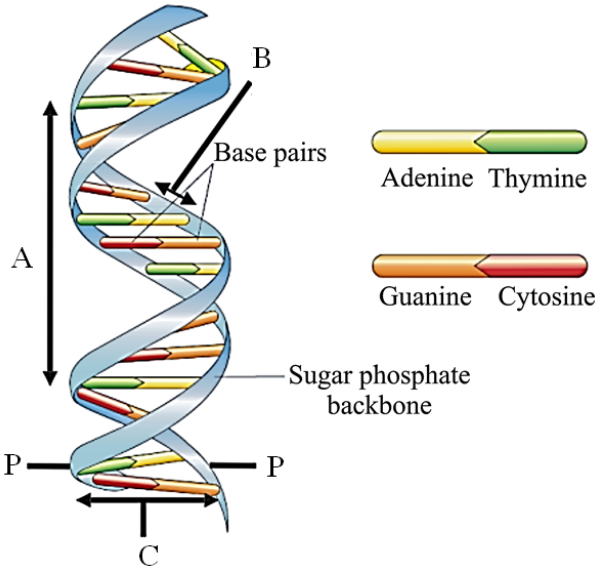
(1) 5'-GACUCGAAC-3'

(2) 5'-CAAGCUCAG-3'

(3) 5-GUUCGAGUC-3'

(4) 3'GUUCGAGUC-5'

24) Given figure represents the DNA double helix model as proposed by Watson and Crick (1953). Select the option that shows correct measurement of A, B and C.



(1) A-3.4 nm, B-0.34 nm, C-2 nm

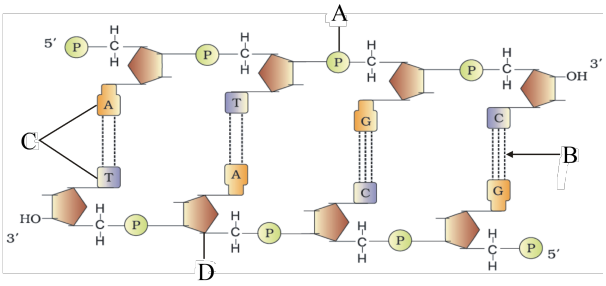
(2) A-34 nm, B-3.4 nm, C-20 nm

(3) A-3.4 Å, B-0.34 Å, C-20 Å

(4) A-34 Å, B-3.4 Å. C-2 Å

25) Given below is the diagrammatic sketch of nucleic acid identify the parts labelled A, B, C & D

and select the right option about it :-



	Part A	Part B	Part C	Part D
(1)	Phospho diester bond	Hydrogen bond	Nitrogenous base	Pentose sugar
(2)	Hydrogen bond	Phospho diester bond	Pentose sugar	Nitrogenous base
(3)	Nitrogenous base	Pentose sugar	Phospho diester bond	Hydrogen bond
(4)	Pentose sugar	Nitrogenous base	Hydrogen bond	Phospho diester bond

(1) 1

(2) 2

(3) 3

(4) 4

26) **Statement-I :-** In eukaryotes negatively charged DNA is wrapped around the positively charged histone octamer to form a structure called nucleosome.
Statement-II :- In prokaryote positively charged DNA is held with some negatively charged proteins in a region termed as nucleoid.

(1) Both statements I and II are correct

(2) Both statement I and II are incorrect

(3) Statement-I is correct and statement-II is incorrect

(4) Statement-I is incorrect and statement-II is correct

27) **Assertion :** RNA is better for the transmission of genetic information.
Reason : RNA can directly code for synthesis of protein.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

28) If a hybrid DNA "N^{15} N^{14}" is allowed to replicate in N^{14} medium, after three generations how many DNA will be having N^{15} :

(1) 3

(2) 4

(3) 8

(4) 1

29) During replication, lagging strand is formed on one of the DNA template :-
How many option are correct related to this
(A) Formation of okazaki fragments.
(B) Direction of fragment formation 5'→ 3'
(C) It is opposite to helix opening.
(D) Need ligase to join.

(1) One

(2) Two

(3) Three

(4) All

30) Lac operon in prokaryotes :-
(a) is regulated negatively
(b) is inducible operon
(c) cannot be regulated positively

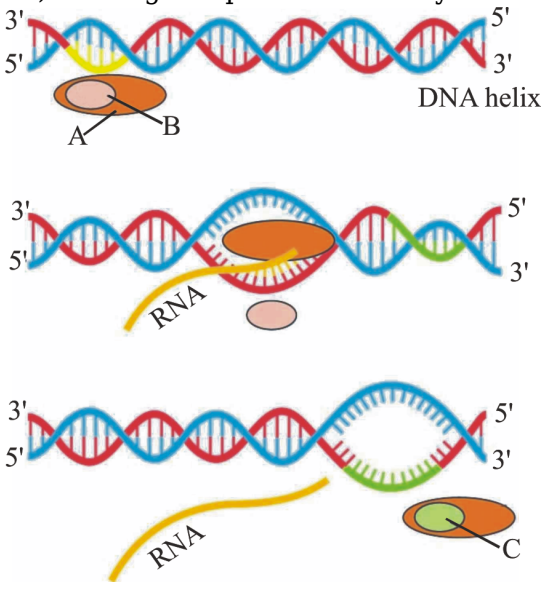
(1) Both (a) and (b) are correct

(2) only (c) is correct

(3) Only (b) is incorrect

(4) Both (b) and (c) are incorrect

31) In the given picture correctly identify A, B and C :-



	A	B	C
(1)	Sigma factor	Rho factor	Stop codon
(2)	RNA polymerase	Sigma factor	Stop codon

(3)	Rho factor	RNA polymerase	Terminator
(4)	RNA polymerase	Sigma factor	Rho factor

(1) 1

(2) 2

(3) 3

(4) 4

32) Match the column:

Column-I		**Column-II**	
(A)	Triplet	(i)	One codon codes for only one amino acid
(B)	Non-ambiguity	(ii)	One amino acid is coded by more than one codon
(C)	Universality	(iii)	A codon is made up of three N_2 bases.
(D)	Degeneracy or redundancy	(iv)	From bacteria to human beings UUU codes for phenyl alanine

(1) A = (iii), B = (ii), C = (iv), D = (i)

(2) A = (iii), B = (i), C = (iv), D = (ii)

(3) A = (iv), B = (i), C = (ii), D = (iii)

(4) A = (iii), B = (iv), C = (i), D = (ii)

33) **Assertion :-** The process of translation is not easy to conceptualise on the basis of complementarity.
Reason :- Neither does any complementarity exist between nucleotides and amino acids, nor could any be drawn theoritically.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

34) Given below are two statements :-
Statement-I : tRNA has an anticodon loop that has bases complementary to the codon of mRNA.
Statement-II : For initiation of transcription, the ribosome binds to the mRNA at the start codon (AUG).
In the light of above statement choose the correct answer from the options given below :

(1) Both Statement-I and Statement-II are incorrect

(2) Statement-I is correct but Statement-II is incorrect

(3) Statement-I is incorrect but Statement-II is correct

(4) Both Statement-I and Statement-II are correct

35) If human cell is cultured in radioactive thymine containing medium then how many chromatids will be radioactive in metaphase :

(1) 46

(2) 92

(3) 2

(4) 23

36) Which of the following enzyme has capability of use a DNA template to catalyse the polymerisation of deoxynucleotides?

(1) DNA dependent RNA polymerase

(2) DNA dependent DNA polymerase

(3) DNA ligase

(4) RNA dependent DNA polymerase

37) Given below are two statements :
Statement-I :- Inheritance of a character is also affected by promoter and regulatory sequences of a structural gene.
Statement-II :- Sometime the regulatory sequences are loosely defined as regulatory genes, even though these sequences do not code for any RNA or protein.

(1) Both Statement-I and Statement-II are correct.

(2) Both Statement-I and Statement-II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

38) Which one can't be used to radioactively label the DNA ?
(a) Radioactive phosphorous ^{32}P
(b) Heavy Nitrogen ^{15}N
(c) Radioactive Sulphur ^{35}S
(d) Radioactive Thymidine

(1) Only b

(2) b,c

(3) b,d

(4) a,b,d

39) Match the columns :-

Column-I		**Column-II**	
(i)	DNA digesting enzyme	(a)	DNAs
(ii)	RNA digesting enzyme	(b)	35S
(iii)	Many DNA molecules	(c)	RNase

(iv)	Radioactive labelled Protein capsule	(d)	DNase

(1) i-a, ii-b, iii-c, iv-d

(2) i-d, ii-c, iii-a, iv-b

(3) i-a, ii-c, iii-d, iv-b

(4) i-c, ii-a, iii-b, iv-d

40) Bacteria which was infected with viruses that had __A__were radioactive and that had__B__were not radioactive. Fill in the blanks A and B.

	A	**B**
(1)	Radioactive Protein	Radioactive Protein DNA
(2)	Radioactive DNA	Radioactive Protein
(3)	Non-Radioactive Protein	Radioactive DNA
(4)	Radioactive Protein	Non-Radioactive DNA

(1) 1

(2) 2

(3) 3

(4) 4

41)

In lac operon non-sense mutation in cistron 'a' leads to synthesis of :

(1) β-galactosidase

(2) Permease

(3) Transacetylase

(4) Both (1) & (2)

42) In a segment of DNA molecule total 4500 N-bases are present then how many sugar molecules will be present in one strand of this DNA segment ?

(1) 9000

(2) 4500

(3) 2250

(4) 450

43) A gene coding for flower colour has 2.8 Kilo base pair. If no. of cytosine molecule in this gene is 1000 then what will be the no. of thymine molecule in this gene ?

(1) 9200

(2) 4600

(3) 3600

(4) 1800

44) "The DNA sequence coding for tRNA and rRNA molecule also define a gene" and segment of DNA coding for a polypeptide could be said as ?

(1) UTR

(2) Cistron

(3) VNTR

(4) Intron

45) Read the following statements :
(i) Eukaryotic mRNA is polycistronic whereas prokaryotic mRNA is monocistronic.
(ii) Prokaryotic mRNA is capless and tail-less whereas eukaryotic mRNA is with cap and tail.
(iii) The precursor of prokaryotic mRNA is hnRNA.
(iv) In replication only a segment of DNA and only one of the strand is copied into RNA.
(v) In eukaryotes, transcription occurs in nucleus and translation in cytoplasm.
Which of the above statements are correct ?

(1) (ii), (iii) and (v)

(2) (iii), (iv) and (v)

(3) (ii) and (v)

(4) (iii) and (iv)

46) Which of the following pair is correctly matched ?

(1) Ribosomal RNA – Carries amino acid to the site of protein synthesis.

(2) Transcription – Process by which protein is synthesised.

(3) Translation – Process by which DNA carries information from the nucleus to ribosomes.

(4) Anticodon – Sequence of a tRNA molecule that binds to the mRNA molecule.

47) In 1953, Watson and Crick proposed DNA double helix model and won the noble prize. Their model of DNA was based on :-
(A) X-ray diffraction studies of DNA done by Wilkins and Franklin.
(B) Chargaff's base equivalence rule.
(C) Griffith's transformation experiment.
(D) Meselson and Stahl's experiment.

(1) A, B and D

(2) A and B

(3) A, B, C and D

(4) B and D

48) In the dsDNA molecule:-

(1) The proportion of Adenine in relation to thymine varies with the organism.

(2) There are two strands which run antiparallel one in 5' →3' direction and other in 3' → 5'.

(3) The total amount of purine nucleotides and pyrimidine nucleotides is not always equal.

(4) There are two strands which run parallel in the 5' →3' direction.

49) During processing of hn-RNA in eukaryotes after transcription, the following event is true :-

(1) In tailing, adenylate residues are added at 5' end in a template dependent manner.

(2) The process of splicing represents the dominance of RNA world.

(3) The split gene arrangement represents probably an advanced feature of genome.

(4) In prokaryotes RNA polymerase-II transcribe hn-RNA.

50) DNA finger printing technology was discovered by Alec Jaffery and this technology uses :-
(a) DNA polymorphism as the basis of identification.
(b) VNTRs, which are proved to be most useful in studies.
(c) Minisatellites having 11-60 bp repeats are used to construct complementary probes.
(d) Gel electrophorosis to amplify the unknown DNA.

(1) All are correct

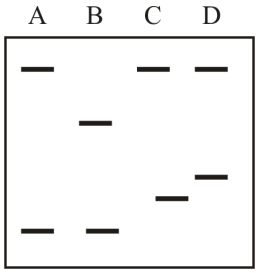
(2) Only (d) is incorrect

(3) Both (a) and (b) are incorrect

(4) Only (c) is incorrect

51)

Out of 4 lane of DNA three lane is of children and one lane is of parent DNA sample. Identify which lane is the lane of parent?



(1) B

(2) A

(3) C

(4) D

52) Read the following statements and select the correct option :-
a. DNA polymorphism is the basis of genetic mapping of human genome as well as DNA finger printing.
b. Minisatellites are also called VNTR.

(1) Only a is correct

(2) Both a & b are incorrect

(3) Only b is correct

(4) Both a & b are correct

53) In humans :-
(I) Non-coding DNA is abundant.
(II) More than 98% of genome codes for protein.

(III) Total number of genes are 3000.
(IV) The function of more than 50% genes are unknown.
Correct options are :-

(1) I, II, III, IV

(2) I, IV

(3) I, III

(4) II, III

54) The satellite DNA is classified into many categories such as micro-satellites, mini-satellites etc. on the basis of :-

(1) Base composition

(2) Length of segment

(3) Number of repetitive units

(4) All the above

55) Which of the following is not used in the DNA finger printing :-

(1) Gel electrophoresis

(2) Northern blotting

(3) PCR

(4) Radioactive probe

56) Mutation in ..... keep on accumulating generation after generation and form one of the basis of DNA polymorphism.

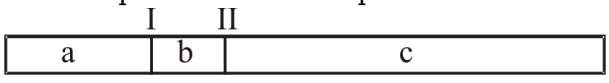
(1) Coding DNA sequence.

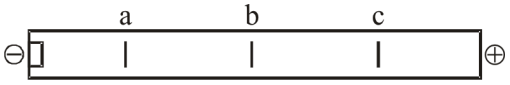
(2) Non-coding DNA sequence.

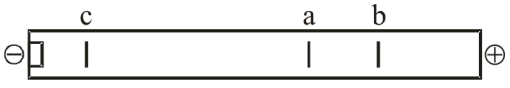
(3) Both coding & non-coding DNA sequence.

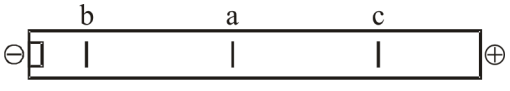
(4) Genetic DNA sequence.

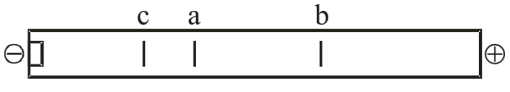
57) A segment of DNA has restriction sites I & II which create restriction fragments a, b and c. Which of the following arrangements produced by gel electrophoresis would represent the separation and correct position of these fragments ?



(1) 

(2) 

(3) 

(4) 

58) During DNA replication one strand is formed continuous and second strand is formed

discontinuous because:-

(1) DNA synthesis is bidirectional.
(2) DNA replication is semiconservative.
(3) DNA chain formation take place in 5' to 3' direction only.
(4) DNA replication starts at ori-site.

59)

Which of the following is not related to stability of DNA in comparison to RNA ?

(1) Deoxyribose sugar
(2) Thymine in place of uracil
(3) Presence of purine
(4) Ability to repair

60) All of the following statements are correct for operon except :

(1) Each operon has its specific operator and specific repressor.
(2) In lac operon lac refers to lactose.
(3) In lac operon i gene refers to inducer.
(4) The i gene code for the repressor of the lac operon.

ANSWER KEYS

BIOLOGY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	**3**	**2**	**4**	**4**	**1**	**4**	**3**	**1**	**2**	**3**	**4**	**1**	**2**	**4**	**4**	**2**	**3**	**1**	**4**	**2**
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	**3**	**3**	**3**	**1**	**1**	**3**	**1**	**4**	**4**	**1**	**4**	**2**	**1**	**2**	**2**	**2**	**1**	**2**	**2**	**2**
Q.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	**4**	**3**	**4**	**2**	**3**	**4**	**2**	**2**	**2**	**2**	**2**	**4**	**2**	**4**	**2**	**2**	**2**	**3**	**3**	**3**

SOLUTIONS

BIOLOGY

1)

Question Explanation: All major steps of protein synthesis require energy.

Concept: Energetics of Translation

Solution :- During Process of Translation energy needed for
(P) Amino acid activation - Attaching amino acid with t- RNA required ATP
(R) T-RNA to P-site - initial t-RNA binding during ignition need of GTP.
(S) t-RNA to A site - subsequent t-RNA binding during elongation need -GTP
⇒ Binding of large ribosomal sub unit does not need energy .

Final Answer : Option (3)

2) **Explain Question :** The question asks to identify what the provided structure represents.

Concept : This question is based on Nucleotides.

Explanation : Since the sugar is ribose and a phosphate group is present, the structure represents a nucleotide of RNA.

Final answer : 2

3)

Question Explanation: DNA nucleosides contain deoxyribose.

Concept: Nucleosides

Solution: A nucleoside of RNA contains ribose sugar. Cytidine, Uridine, and Guanosine are all RNA nucleosides. Deoxyguanosine contains the sugar deoxyribose, making it a nucleoside of DNA, not RNA.

Final Answer: Option(4). Deoxyguanosine

4)

NCERT (XIth) Pg. # 96, Para # 3

5) NCERT (XIth) Pg. # 97, Para # 3

6)

Question Explanation: Phosphodiester bonds form the backbone of a DNA strand.

Concept: Polynucleotide Structure.

Solution: In a single strand of DNA, the successive nucleotides are covalently linked together by phosphodiester bonds. This bond connects the 3' carbon of one sugar molecule to the 5' carbon of the next sugar molecule via a phosphate group, forming the strong sugar-phosphate backbone.

Final Answer: Option (4). Phosphodiester bond

7)

Question explanation: The question asks which molecule allolactose binds to in order to activate (turn on) the lac operon.

Concept: Lac operon

Solution: Allolactose → binds to repressor protein → repressor inactive → operon ON.

Final answer: Repressor protein

8) NCERT Pg # 120

9)

NCERT XII, Pg # 120

10) NCERT (XII) Pg. # 99

11) Correct Answer: (4)

12) Correct Answer: (1)

13) Correct Answer: (2)

14) Correct Answer: (4)

15)

NCERT XII, Pg # 99

16) Correct Answer: (2)

17) NCERT (XII) ; Page No. # 81

18) Correct Answer: (1)

19) NCERT-XII, Pg. # 93, 94

20) NCERT Pg. # 96 (Structure of Polynucleotide Chain)

21)

NCERT XII, Pg # 84-85

22) NCERT Pg. # 86 (Properties of Genetic Material)

23)

Question explanation: The question asks to determine the mRNA sequence from the coding strand of DNA

Concept :
Transcription

Explanation :
The coding strand of DNA: 5'-GTTCGAGTC-3'
During transcription, mRNA is complementary to the template strand and has the same sequence as the coding strand, except T → U.

So, mRNA sequence: 5'-GUUC GAG UC-3

Final answer : 5-GUUCGAGUC-3'

24) NCERT Pg # 98

25) **Explain Question :**This question is about identifying the different components and bonds within a diagrammatic sketch of a DNA double helix.
Concept : DNA Structure And Features.
Explanation :
Part A: Points to the bond that connects the phosphate group (P) of one nucleotide to the pentose sugar of the adjacent nucleotide within the same strand. This bond is the Phosphodiester bond.
Part B: Points to the bonds linking the complementary nitrogenous bases (C and G, or A and T) across the two strands. These bonds hold the two helix strands together. This is the Hydrogen bond.
Part C: Points to the entire heterocyclic ring structure, such as 'A' (Adenine) or 'T' (Thymine), that projects inward from the sugar-phosphate backbone. This is the Nitrogenous base.
Part D: Points to the five-carbon ring structure (pentagon shape in the diagram) in the backbone, which is attached to the phosphate group and the nitrogenous base. This is the Pentose sugar (Deoxyribose).
Final answer : 1

26)

NCERT, Pg. # 99

27) NCERT Pg#103

28)

Question Explanation : Meselson–Stahl experiment (semiconservative replication)

Concept: DNA replication : Semiconservative model

Solution:

A. Start: 1 hybrid (N15–N14)

B. After 1st generation → 1 hybrid + 1 light

C. After 2nd generation → 1 hybrid + 3 light

D. After 3rd generation → 1 hybrid + 7 light

So, only **1 DNA** still has N15.

Final answer: Option **(4)**

29)

Explanation:
The lagging strand is synthesized discontinuously as Okazaki fragments (A), each formed in the 5' to 3' direction (B), with the overall synthesis occurring opposite to the direction of helix unwinding (C). These fragments are subsequently joined together by DNA ligase (D).

Correct Answer: 4. All

30)

Question explanation: The question tests understanding of lac operon regulation in prokaryotes.

Concept: Lac operon

Solution: The lac operon is negatively regulated by a repressor protein that blocks

transcription in the absence of lactose → correct

It is an inducible operon because lactose (via allolactose) induces its expression → correct

It can also be positively regulated by cAMP–CAP complex when glucose is low → so “cannot be regulated positively” is - incorrect

Thus statement A and B are correct

Final answer: Both (a) and (b) are correct

31)

Question Explanation: The question asks for the identification of A, B, and C in the prokaryotic transcription diagram.
Concept: Transcription in Prokaryotes
Solution:
RNA polymerase (A) binds to DNA with the help of sigma factor (B) at the promoter.
Sigma factor assists RNA polymerase in recognizing the promoter.
Rho factor (C) binds RNA to terminate transcription by releasing the RNA from RNA polymerase.
Final Answer: Option (4)

32) NCERT XII Pg. 112

33) **Explain Question :**The question asks to evaluate the Assertion and Reason

Concept : Translation

Explanation : Assertion: True — translation isn’t based on complementarity.

Reason: True — no direct complementarity exists; tRNA mediates the code.

The Reason directly explains why translation cannot be conceptualized by complementarity

Final answer : Option (1)

34) **Explain Question –**
The question asks us to evaluate two statements about tRNA function and transcription/translation initiation, and then choose the correct option.

Concept –
Translation (role of tRNA, start codon AUG) vs Transcription (initiation by RNA polymerase at promoter).

Solution –
Statement-I: Correct. tRNA has an anticodon loop with bases complementary to the codon on mRNA.
Statement-II: Incorrect. Ribosome binds to mRNA at the start codon AUG during translation, not transcription. Transcription initiation involves RNA polymerase binding to the promoter

region, not ribosome.
Thus, only Statement-I is correct.

Final Answer : Option 2

35)

NCERT Pg # 106

36) **Explain Question :**The question asks which enzyme can use a DNA template to polymerise deoxynucleotides (i.e., make DNA)

Concept : DNA replication enzymes

Explanation : DNA-dependent DNA polymerase uses an existing DNA strand as a template to add deoxynucleotides and synthesize a new DNA strand.

Final answer : Option (2)

37) NCERT-XII Pg. # 93

38)

Explain Question :This question asks which substances cannot be used to label DNA radioactively in experiments

Concept : Search Of Genetic Material

Explanation :
^{15}N (heavy nitrogen) → generally used in density experiments (Meselson-Stahl) but not a radioactive label → Incorrect cannot be used to radioactively label DNA

^{35}S (radioactive sulfur) → labels protein, not DNA → Incorrect cannot label DNA

Final answer :Option (2)

39)

Explain Question :This question asks to correctly match enzymes, molecules, and labels with their functions or identities in classic DNA experiments.

Concept : Search Of Genetic Material

Explanation :
i. DNA digesting enzyme → DNase (d)
ii. Griffith's Transforming Principle → DNA (c)
iii. Many DNA molecules → DNA (a)
iv. Radioactive labelled Protein capsule → ^{35}S (b)

Final Answer: Option (2)

40)

Explain Question :This question asks which viral component (DNA or protein) entered bacteria during the Hershey-Chase experiment, making the bacteria radioactive.

Concept : Search Of Genetic Material

Explanation :
Bacteria infected with viruses having radioactive DNA became radioactive.

Bacteria infected with viruses having radioactive protein did not become radioactive.

Final answer :Option (2)

41)

From NCERT
Correct Ans. # (4)

42) NCERT-XI Pg. # 151

43)

Question Explanation: The question asks the no. of thymine molecules in the given context.

Concept: DNA Structure.

Solution: Number of cytosine (C) molecules = 1000

By base-pairing rule:

A. C pairs with G
   ⇒ Number of guanine (G) = 1000

So together, C+G=2000 nucleotides (or 1000 base pairs of C–G).

Total base pairs = 2800

Remaining base pairs = 2800−1000=1800 base pairs of A–T.

So, the correct answer is: 4. 1800

Final answer: Option (4)

44)

Explain Question: The question distinguishes between different types of DNA segments — those coding for RNAs (tRNA, rRNA) and those coding for proteins (polypeptides).

Concept: Transcription unit
Solution: Segment of DNA coding for a polypeptide = Cistron
Final answer: Cistron

45)

Question explanation: The question asks to find out which of the given statements are correct.

Concept: split gene

Solution:

The Correct statements are:

(ii) Prokaryotic mRNA is capless and tail-less whereas eukaryotic mRNA is with cap and tail.

(v) In eukaryotes, transcription occurs in the nucleus and translation in cytoplasm.

So, the correct answer is: 3. (ii) and (v)

Final answer: (ii) and (v)

46) **Concept :-** Anticodon loop of t- RNNA have just complimentary sequence of m-RNA's codon.
Solution :- Anticodon - site of a t- RNA molecule that bind to m-RNA molecule is the correct match from given.
bcz - The anticodon is a specific three nucleotide sequence of t- RNA molecule. This sequence can base pair with a complimentary three nucleotide sequence called codon on the m- RNA molecule This Pairing is crucial for ensuring the correct amino awed is added during translation .
Final Answer: Option 4

47) NCERT (XII) Pg # 96

48)

From NCERT
Correct Ans. # (2)

49)

From NCERT
Correct Ans. # (2)

50) NCERT-XII Pg. # 121-122

51)

NCERT XII Pg # 106(E)

52)

Question Explanation: DNA polymorphism and its terminology.

Concept: DNA Fingerprinting

Solution: Both statements are correct. Statement 'a' is true because the variation in DNA sequences (polymorphism) is the fundamental principle behind creating genetic maps and performing DNA fingerprinting. Statement 'b' is also true; minisatellites are regions of Variable Number of Tandem Repeats (VNTRs), and the terms are often used interchangeably in this context.

Final Answer: 4. Both a & b are correct

53) NCERT (XIIth) Pg. # 120 (Para 6.9.1)

54) NCERT-XII, Pg. # 105

55)

Question Explanation: A technique not used in DNA fingerprinting.
Concept: DNA Fingerprinting

Solution: The standard DNA fingerprinting technique involves DNA digestion, gel electrophoresis, Southern blotting, and hybridization with a probe. Northern blotting is a similar molecular technique, but it is used to detect specific RNA molecules from a sample, not DNA.
Final Answer: 2. Northern blotting

56) NCERT XII Pg.# 122

57)

Question Explanation: Predicting the result of gel electrophoresis.
Concept: Gel Electrophoresis

Solution: Gel electrophoresis separates DNA fragments based on size; smaller fragments migrate faster and further towards the positive electrode. From the diagram of the original DNA, fragment 'b' is the smallest, 'a' is medium, and 'c' is the largest. Therefore, on the gel, 'c' will be closest to the start (negative electrode), 'a' will be in the middle, and 'b' will be the farthest. This pattern is shown in option 4.
Final Answer: 4.

58)

NCERT XII Pg. # 106

59)

From NCERT
Correct Ans. # (3)

60) NCERT-XII Pg. # 100,101